

S.Y. B. Tech. Academic Year 2026-27 Semester: III
CSE20350/AID20210/CLD 20020: Data Structures Laboratory

WRITE UP

Experiment Number: 02
TITLE: Sparse Matrix Operations

PROBLEM STATEMENT:

Write a C program for sparse matrix realization and operations on it- Transpose, Fast Transpose

OBJECTIVE:

1. To Study the concept of sparse matrix, how it is stored and displayed.
2. To understand the implementation of sparse matrix operations – Simple Transpose, Fast Transpose.

IMPLEMENTATION:

- **PLATFORM:**

- 64-bit Open source Linux or its derivatives.
- Open Source C Programming tool like gcc/Eclipse Editor.

- **PSEUDO CODE:** *//To be Written by Students*

Write pseudo code for input matrix to compact form, Simple Transpose and Fast Transpose.

- **TIME COMPLEXITY:** *//To be Written by Students*

Find out the time complexity of Simple Transpose and Fast Transpose.

- **CONCLUSION:**

Thus, implemented sparse matrix Operations assignment. This System is able to perform different operations on sparse matrices such as simple and fast transpose and their time complexities.

- **FAQs**

//To be Written by Students

- What is a sparse matrix? List the applications?
- Represent sparse matrices with suitable data structures? Explain with an example simple and fast transpose?
- Find out the addition of two sparse matrices in triplet form and also find Simple and Fast transpose?

M1 =

4	5	6
0	3	5
1	3	8
1	4	45
2	3	4
3	2	45
4	1	2

M2 =

4	5	6
0	3	7
0	4	6
1	4	4
2	1	8
3	2	45
4	4	21

- **PRACTICE ASSIGNMENTS**

- Write a program to perform sparse matrix operations – addition and multiplications.